

Operations Research (MBA204 / 22MBA24)

Repeated Question Analysis — June/July 2023, 2024 & 2025

This document compares the three VTU/CBCS Second Semester MBA question papers for **Operations Research** — **June/July 2023**, **June/July 2024**, and **June/July 2025** — and identifies questions/topics that repeat across years. Repeats are organised module-wise (Module 1 to Module 6, per the official syllabus), with the marks and year(s) each question repeated from. A separate section at the end lists the most repeated **practical / numerical** question types.

Legend: **EXACT REPEAT** = same/near-identical wording or data across years **REPEATED THEME** = same topic/concept asked again (numbers or phrasing may differ)

MODULE 1: Introduction to Operations Research

- Define / What is Operations Research?

3 Marks — 2023 (Q1a) | 3 Marks — 2024 (Q1a) | 7 Marks — 2025 (Q2b, combined with 'phases') **REPEATED THEME**

Core definition question — asked in every single year, either standalone (3M) or combined with phases (7M).

- Applications of Linear Programming Problem (LPP) in management

3 Marks — 2023 (Q2a) | 7 Marks — 2024 (Q1b) **REPEATED THEME**

- Scope of Operations Research

7 Marks — 2023 (Q2b)

Appears once in this set (not repeated across years) — but is a core Module-1 topic, worth preparing.

- Historical development / evolution of OR

3 Marks — 2025 (Q1a)

Appears once in this set.

MODULE 2: Linear Programming

- Formulate a real-world scenario as an LPP (products/raw material, investment portfolio, etc.)

7 Marks — 2025 (Q1b) — two products, raw material & processing time | 7 Marks — 2023 (Q1b) — two models, grinding & polishing | 10 Marks — 2024 (Q1c) — mutual fund investment allocation **REPEATED THEME**

LPP formulation from a word problem is asked every single year without exception — guaranteed 7–10 mark question.

- Solve LPP by Graphical Method — Min $z = 20x + 10y$, s.t. $x + 2y \leq 40$, $3x + y \geq 30$, $4x + 3y \geq 60$, $x, y \geq 0$

10 Marks — 2023 (Q3c) | 10 Marks — 2024 (Q2c) **EXACT REPEAT**

Identical problem (only variable names x, y vs x_1, x_2 differ) — set in back-to-back years. High chance of reappearing.

- Solve LPP by Graphical Method (general — maximize/minimize with 2 constraints)

10 Marks — 2025 (Q1c) — Maximize $Z = 6x_1 + 8x_2$ **REPEATED THEME**

Graphical-method LPP appears in all 3 papers (see exact repeat above for 2023/2024); 2025 uses a different maximization problem.

- Limitations of LPP

3 Marks — 2024 (Q2a)

Appears once in this set.

- Limitations of graphical method

3 Marks — 2024 (Q5a)

Appears once in this set.

MODULE 3: Decision Theory, Job Sequencing

- Basic assumptions of sequencing problems

3 Marks — 2025 (Q2a) | 3 Marks — 2024 (Q7a) — 'any three assumptions' **REPEATED THEME**

- Importance / definition of (job) sequencing

3 Marks — 2025 (Q6a) | 3 Marks — 2023 (Q6a) — 'Why is job sequencing important?' **REPEATED THEME**

- Steps involved in the Decision-Making process

7 Marks — 2023 (Q6b) | 7 Marks — 2024 (Q2b) **REPEATED THEME**

- Sequencing numerical — process 'n' jobs through 2 or 3 machines; find optimal sequence, total elapsed time & idle time

10 Marks — 2025 (Q2c) — 5 jobs, 2 machines | 10 Marks — 2025 (Q6c) — 3 jobs, 3 machines | 10 Marks — 2023 (Q1c) — 4 jobs, 3 machines | 10 Marks — 2024 (Q4c) — 5 jobs, 3 machines **REPEATED THEME**

The single most frequently repeated numerical in this module — appears multiple times, almost always for 10 marks.

- Decision theory — short note / criteria for decision-making under uncertainty (Maximin, Minimax)
3 Marks — 2025 (Q5a) | 7 Marks — 2023 (Q5b) | 7 Marks — 2025 (Q6b) — Max-Min/Min-Max with example **REPEATED THEME**

MODULE 4: Transportation Problems

- What is an unbalanced transportation problem?
3 Marks — 2025 (Q3a) | 3 Marks — 2024 (Q3a) **REPEATED THEME**
- What is Degeneracy in a transportation problem?
3 Marks — 2025 (Q4a) | 3 Marks — 2023 (Q7a) **REPEATED THEME**
- Find the Initial Basic Feasible Solution using North-West Corner Rule (NWCR/NWCM)
7 Marks — 2025 (Q3b) | 7 Marks — 2024 (Q3b) **REPEATED THEME**
- Find the Initial Basic Feasible Solution using Vogel's Approximation Method (VAM)
7 Marks — 2025 (Q5b) | 10 Marks — 2024 (Q3c) **REPEATED THEME**
VAM numericals appear in 2 of the 3 years; combine with MODI for optimality as it's often asked together (see Q.8 case studies below).
- Solve a transportation problem by Least Cost Method (to maximize profit)
10 Marks — 2023 (Q2c)
Appears once in this set.
- Check optimality of a given transportation solution (MODI method)
10 Marks — 2024 (Q7c) — given solution table, justify optimality | 20 Marks — 2023 (Q8 case study) — VAM + MODI + alternate optimal solution **REPEATED THEME**
- Compulsory Q.8 large case study — full transportation problem (formulate, solve via VAM, check optimality via MODI, find minimum cost/schedule)
20 Marks — 2025 (Q8) — Larsen & Toubro construction fill supply | 20 Marks — 2023 (Q8) — factories/warehouse capacity-demand table **REPEATED THEME**
In 2 of 3 years, the compulsory 20-mark Q.8 was a full transportation case study. (In 2024, Q.8 instead combined a network-diagram float question with an assignment problem — see Module 5 & 6.)

MODULE 5: Theory of Games & Assignment Problem

- Solve a game using the Principle of Dominance
7 Marks — 2025 (Q4b) | 7 Marks — 2023 (Q3b) — payoff matrix of Company X/Y **REPEATED THEME**
- Solve a game using the Graphical Method ($2 \times n$ game)
10 Marks — 2025 (Q4c) — 2×3 game | 10 Marks — 2023 (Q6c) | 7 Marks — 2024 (Q5b) **REPEATED THEME**
Graphical-method game problems appear in ALL THREE years — one of the most reliably repeated numericals in the paper.
- Define terms in Game Theory (Pay-off matrix, Pure/Mixed Strategy, Value of Game, Fair Game) / Saddle Point
10 Marks — 2025 (Q5c) — define 5 terms | 3 Marks — 2024 (Q4a) — Saddle point in game theory **REPEATED THEME**
- What is an Assignment Problem?
3 Marks — 2023 (Q3a)
Appears once in this set.
- Solve an Assignment Problem by Hungarian Method (minimize cost / minimize time)
10 Marks — 2025 (Q3c) | 7 Marks — 2023 (Q7b) | 7 Marks — 2024 (Q6b) — assign programmers to programs **REPEATED THEME**
Assignment-problem numericals repeat in every year — a near-certain question.

- Optimal assignment schedule with restrictions (a machine/person cannot be assigned to a specific task)

10 Marks — 2024 (Q8b)

Appears once in this set, but is a natural variant of the Hungarian-method question above.

- Solve a game problem by the Algebraic Method

10 Marks — 2024 (Q6c)

Appears once in this set.

MODULE 6: Project Management — PERT & CPM

- Differentiate / Distinguish between PERT and CPM

7 Marks — 2025 (Q7b) | **7 Marks** — 2023 (Q4b) | **7 Marks** — 2024 (Q7b) **REPEATED THEME**

Asked in ALL THREE years without fail, always for 7 marks — the single most reliably repeated Module-6 theory question.

- Construct network diagram and find Critical Path from an activity/predecessor/duration table

10 Marks — 2025 (Q7c) — arrow diagram + critical path + project duration | **10 Marks** — 2023 (Q5c) — precedence relation + critical path + floats | **10 Marks** — 2024 (Q5c) — network diagram + critical path + completion time | **10 Marks** — 2024 (Q8a) — Es, Ef, Ls, Lf & total float **REPEATED THEME**

The core CPM numerical — appears at least once (often twice) in every paper, always for 10 marks.

- Phases of Project Management

3 Marks — 2025 (Q7a)

Appears once in this set; related standalone topics in other years include 'Work-Breakdown Structure' (2023 Q4a) and 'Project Control Phase' (2023 Q5a).

- PERT — three time estimates (optimistic, most likely, pessimistic): find expected time, critical path & completion probability

10 Marks — 2023 (Q7c)

Appears once in this set — but is an important, high-mark numerical worth practicing given it combines Module-6 statistics with CPM.

MOST REPEATED PRACTICAL / NUMERICAL QUESTIONS

The table below ranks the numerical/problem-solving question **types** that repeat most often across the three papers (2023, 2024, 2025), with the specific questions asked each year.

1. Network Diagram & Critical Path Method (CPM) numerical

- 2025 Q7c — arrow diagram, critical path, project duration (10M)
- 2023 Q5c — network diagram, critical path, total floats (10M)
- 2024 Q5c — network diagram, critical path, completion time (10M)
- 2024 Q8a — Es, Ef, Ls, Lf and total float (10M)

Appeared in all 3 years, sometimes twice in the same paper — the single most reliable practical question. Master float calculations (Es/Ef/Ls/Lf) and critical-path identification.

2. Game Theory — Graphical Method for 2×n games

- 2025 Q4c — 2×3 game, graphical method (10M)
- 2023 Q6c — solve game graphically (10M)
- 2024 Q5b — solve game using graphical approach (7M)

Appeared in all 3 years. Practice plotting payoff lines and identifying the maximin point.

3. Job Sequencing ('n' jobs through 2 or 3 machines)

- 2025 Q2c — 5 jobs, 2 machines (10M)
- 2025 Q6c — 3 operations, 3 machines (10M)
- 2023 Q1c — 4 jobs, 3 machines (shaping-drilling-tapping) (10M)
- 2024 Q4c — 5 jobs, 3 machines (10M)

Appeared in all 3 years, sometimes twice in one paper. Always 10 marks — practice both 2-machine and 3-machine cases, plus finding idle time.

4. Assignment Problem — Hungarian Method

- 2025 Q3c — assign 3 tasks to 3 persons, minimize cost (10M)
- 2023 Q7b — solve the assignment problem (7M)
- 2024 Q6b — assign programmers to programs, minimize time (7M)
- 2024 Q8b — assignment with restrictions (specific cells barred) (10M)

Appeared in all 3 years, sometimes twice. Practice the standard Hungarian method plus the 'restricted cell' variant.

5. LPP — Graphical Method

- 2025 Q1c — Maximize $Z=6x_1+8x_2$, s.t. $x_1+x_2\leq 450$, $2x_1+x_2\leq 600$ (10M)
- 2023 Q3c — Minimize $z=20x+10y$, s.t. $x+2y\leq 40$, $3x+y\geq 30$, $4x+3y\geq 60$ (10M)
- 2024 Q2c — Minimize $z=20x_1+10x_2$, s.t. $x_1+2x_2\leq 40$, $3x_1+x_2\geq 30$, $4x_1+3x_2\geq 60$ (10M) — IDENTICAL to 2023 Q3c

Appeared in all 3 years; 2023 and 2024 used the exact same numbers. Strong candidate to reappear.

6. LPP Formulation (from a word problem)

- 2025 Q1b — two products, raw material & processing-time constraints (7M)
- 2023 Q1b — two models, grinding & polishing-hour constraints (7M)
- 2024 Q1c — mutual fund investment allocation across 4 instruments (10M)

Appeared in all 3 years without exception — guaranteed question, though the scenario/numbers change each time.

7. Transportation Problem — Initial Feasible Solution (NWCR / VAM / Least Cost)

- 2025 Q3b — NWCR (7M); 2025 Q5b — VAM (7M)
- 2023 Q2c — Least Cost Method, maximize profit (10M)
- 2024 Q3b — NWCM (7M); 2024 Q3c — VAM (10M)

NWCR and VAM each appear in 2 of the 3 years. Be comfortable with all three initial-solution methods (NWCR, Least Cost, VAM).

8. Compulsory Q.8 — Full Transportation Case Study (VAM + MODI optimality)

- 2025 Q8 — Larsen & Toubro construction, minimize transport cost across 3 sources/4 destinations (20M)
- 2023 Q8 — factories-to-warehouse capacity/demand table, VAM + MODI + alternate optimal check (20M)

The compulsory Q.8 was a transportation case study in 2 of the 3 years — high-value (20M) and worth prioritizing.

9. Game Theory — Principle of Dominance

- 2025 Q4b — solve using concept of dominance (7M)
- 2023 Q3b — payoff matrix, solve by dominance (7M)

Appeared in 2 of 3 years.